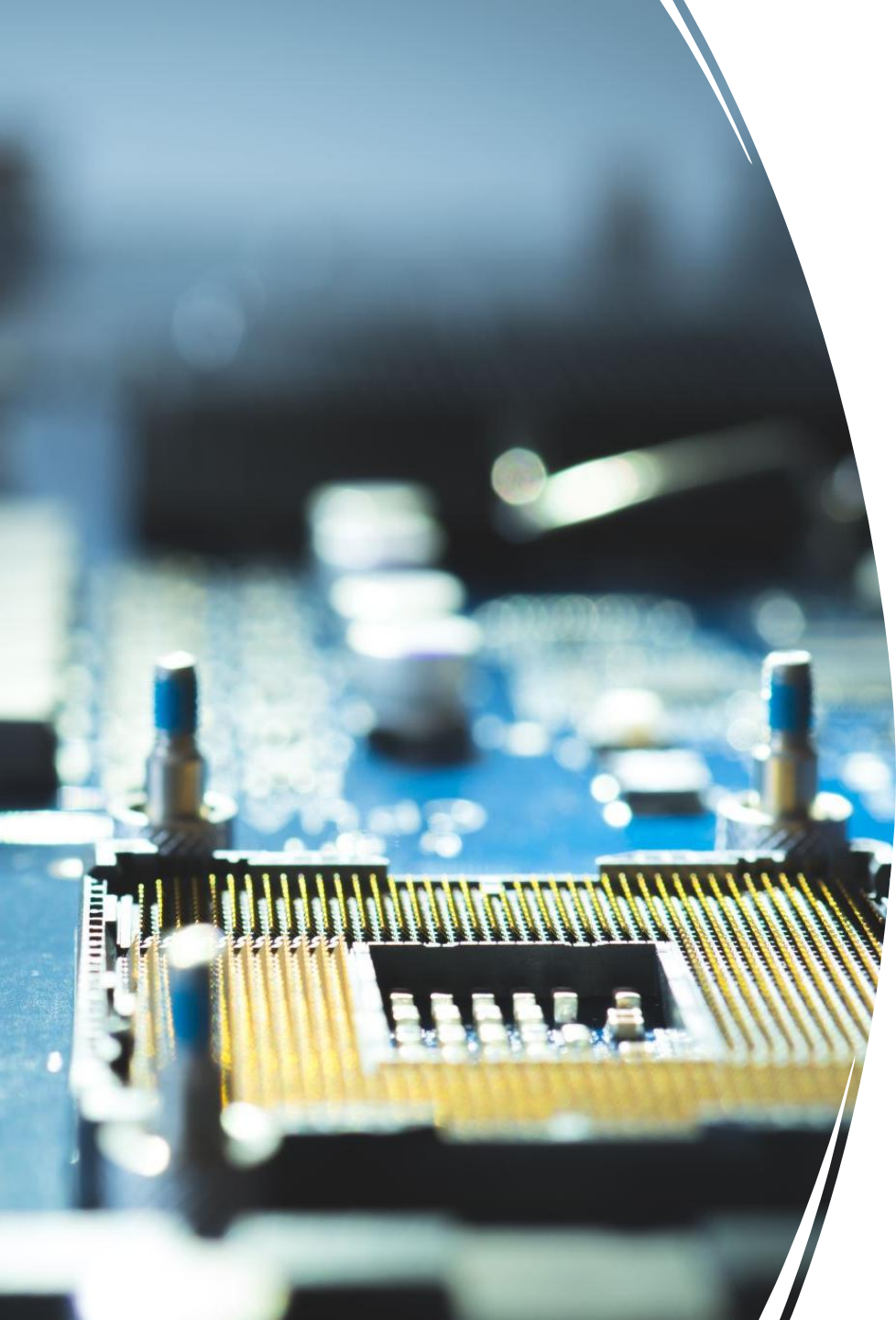




HIA – Hardware Intelligence Assistant

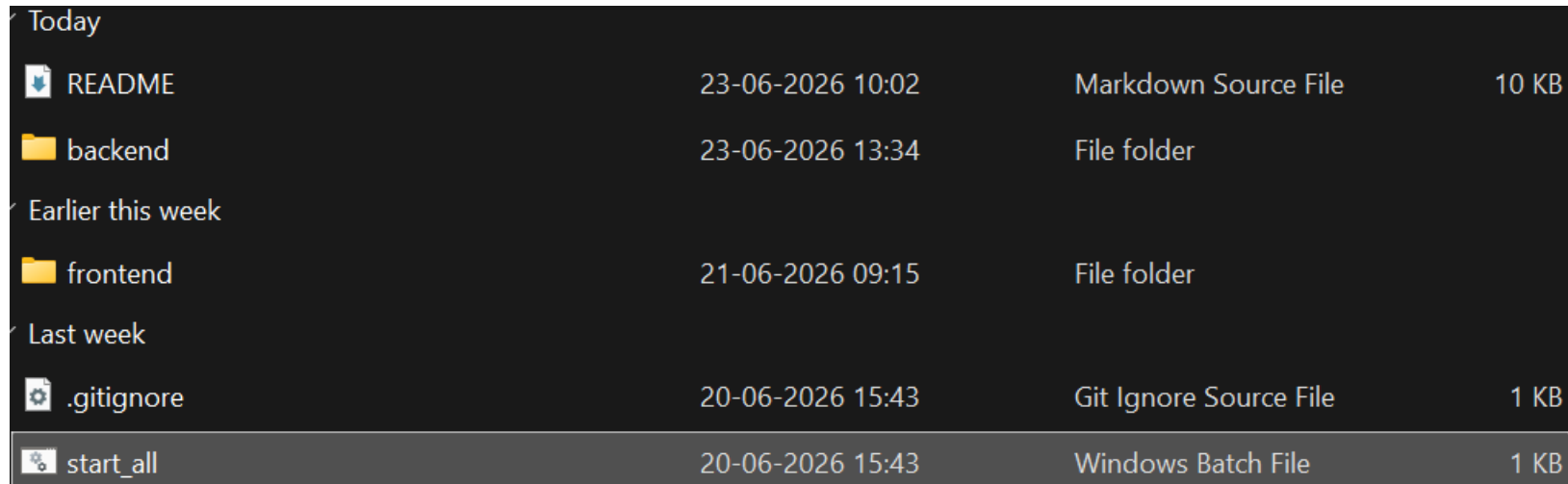
An AI-Powered Hardware Analysis and Diagnostic Platform
Developed for Flowzint AI Hackathon 2026

Individual Participation
Name: Chinmay U Sidnal

Unzip the folder

 HIA_FI_Hackathon	23-06-2026 10:20	Compressed (zipped)...	4,57,625 KB
 HIA_FI_Hackathon	23-06-2026 15:54	File folder	

Open the file start_all



Today				
	README	23-06-2026 10:02	Markdown Source File	10 KB
	backend	23-06-2026 13:34	File folder	
Earlier this week				
	frontend	21-06-2026 09:15	File folder	
Last week				
	.gitignore	20-06-2026 15:43	Git Ignore Source File	1 KB
	start_all	20-06-2026 15:43	Windows Batch File	1 KB

THE INITIAL STARTUP WILL TAKE MORE TIME

IT WILL DOWNLOAD THE NECESSARY REQRUMENTS

NOTE: **qwen2.5vl.latest** is around 6gb , and is fast with dedicated gpus.

This is the first cmd window that will download the necessary applications.

It downloads the applications that are not present.

The first startup will take around 20mins depending upon the specs of device (dedicated gpu is faster).

```
HIA - Hardware Intelligence Assistant
=====

HIA runs completely on your local machine.

First startup may download:

Python dependencies:
- FastAPI
- PyTorch
- Ultralytics YOLOv8
- EasyOCR
- OpenCV
- SQLAlchemy
- Other required packages

Approximate dependency size:
2 GB to 4 GB

AI Model:
- qwen2.5vl:latest
  Approximate size: 6 GB

Total first-time download:
Approximately 8 GB to 10 GB

Already installed components
will NOT be downloaded again.
=====
```

Wait till you see
startup
complete in
backend

```
Requirement already satisfied: jinja2 in .\venv\Lib\site-packages (from torch>=1.8.0->ultralytics>=8.2.0->-r requirements.txt (line 25)) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in .\venv\Lib\site-packages (from torch>=1.8.0->ultralytics>=8.2.0->-r requirements.txt (line 25)) (2026.6.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in .\venv\Lib\site-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics>=8.2.0->-r requirements.txt (line 25)) (1.3.0)
Requirement already satisfied: httptools>=0.8.0 in .\venv\Lib\site-packages (from uvicorn[standard]>=0.30.0->-r requirements.txt (line 3)) (0.8.0)
Requirement already satisfied: watchfiles>=0.20 in .\venv\Lib\site-packages (from uvicorn[standard]>=0.30.0->-r requirements.txt (line 3)) (1.2.0)
Requirement already satisfied: websockets>=10.4 in .\venv\Lib\site-packages (from uvicorn[standard]>=0.30.0->-r requirements.txt (line 3)) (16.0)
Requirement already satisfied: MarkupSafe>=2.0 in .\venv\Lib\site-packages (from jinja2->torch>=1.8.0->ultralytics>=8.2.0->-r requirements.txt (line 25)) (3.0.3)
Requirement already satisfied: imageio!=2.35.0,>=2.33 in .\venv\Lib\site-packages (from scikit-image->easyocr>=1.7.1->-r requirements.txt (line 29)) (2.37.3)
Requirement already satisfied: tifffile>=2022.8.12 in .\venv\Lib\site-packages (from scikit-image->easyocr>=1.7.1->-r requirements.txt (line 29)) (2026.6.1)
Requirement already satisfied: lazy-loader>=0.4 in .\venv\Lib\site-packages (from scikit-image->easyocr>=1.7.1->-r requirements.txt (line 29)) (0.5)

=====
Starting backend at http://localhost:8000
API docs available at http://localhost:8000/docs
Press CTRL+C to stop.
=====

INFO: Will watch for changes in these directories: ['C:\\Users\\Chinmay U Sidnal\\Downloads\\hardware-diagnostic-assistant_4\\hardware-diagnostic-assistant\\backend']
Using CPU. Note: This module is much faster with a GPU.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Started reloader process [25340] using WatchFiles
Using CPU. Note: This module is much faster with a GPU.
INFO: Started server process [40720]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

```
=====
Starting backend at http://localhost:8000
API docs available at http://localhost:8000/docs
Press CTRL+C to stop.
=====
```

```
INFO:      Will watch for changes in these directories: ['C:\\Users\\Chir
U Sidnal\\Downloads\\hardware-diagnostic-assistant_4\\hardware-diagnosti
sistant\\backend']
Using CPU. Note: This module is much faster with a GPU.
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:      Started reloader process [25340] using WatchFiles
Using CPU. Note: This module is much faster with a GPU.
INFO:      Started server process [40720]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
```

Create your acc.
For demo use these

name: User12
password: User12



Hardware Diagnostic Assistant

AI-powered analysis for embedded systems & electronics

Sign In

Create Account

Username

your_username

Email

you@example.com

Password

At least 6 characters

Create Account

Runs entirely on your machine via a local Ollama model.

📄

HDA

+

New Chat

🔍

Search chats

💬

Chat

📷

Vision Lab

No chats yet. Start a new one!

U

User_12

User.12@gmail.com

Log Out

Diagnostic Chat

Streaming · local Ollama model

⚡ Hardware Engineer Mode

⚡

Hardware Engineer Mode

Describe a circuit, microcontroller, or hardware issue and I'll help you debug it — PCB repair, IC identification, embedded systems, power electronics, and more.

ESP32 brownout on WiFi connect

Identify an unmarked IC

PWM vs analog for motor control

📎

Describe the issue you're debugging... e.g. 'ESP32 keeps brownout-resetting on WiFi

🔊

📤

Enter to send · Shift+Enter for a new line

📁 HDA

+ New Chat

🔍 Search chats

💬 Chat

📷 Vision Lab

No chats yet. Start a new one!

U User_12

User.12@gmail.com

Log Out

Vision Lab

YOLO detection · OCR · AI analysis

⚡ Hardware Engineer Mode

IMAGE SOURCE

↑

Drag & drop, or [browse](#)

JPG, PNG, WEBP supported

📷 Use Webcam Instead

Upload or capture an image to begin analysis.

Results will appear here after analysis.

RECENT ANALYSES

No analyses yet.



HDA

+ New Chat

Search chats

Chat

Vision Lab

No chats yet. Start a new one!



User_12
User.12@gmail.com

Log Out

Vision Lab

YOLO detection · OCR · AI analysis



Drag & drop, or [browse](#)
JPG, PNG, WEBP supported

Use Webcam Instead

Selected Image



Run Diagnostic Analysis



RECENT ANALYSES



23/06/2026



23/06/2026



23/06/2026



23/06/2026



23/06/2026

Hardware Engineer Mode

Resistor

98.5%

Resistor

97.8%

Resistor

96.9%

Resistor

96.8%

Resistor

96.7%

Resistor

96.2%

OCR RESULTS

No text detected in this image.

AI ANALYSIS & REPAIR SUGGESTIONS

Based on the photo, I detect what appears to be a resistor. The lack of readable labels or markings suggests that it may be an unmarked or generic part. The presence of a resistor in this context implies that it's likely serving as a component in a circuit, possibly influencing voltage levels or current flow.

Without further information, it's challenging to

